Atiyah–Macdonald, Chapter 2: Modules

Solutions to the exercises

Conventions. Rings are commutative with 1; directed sets are non-empty; $\otimes = \otimes_A$ unless a subscript says otherwise; Im is the image. From the text I use freely: Cayley–Hamilton for finitely generated modules (2.4), Nakayama's lemma (2.6, 2.7), the canonical isomorphisms of 2.14 (commutativity, associativity, distributivity over $\oplus$, $A \otimes M \cong M$) and 2.15, right exactness of $\otimes$ (2.18), $\mathrm{Hom}(M \otimes N, P) \cong \mathrm{Hom}(M, \mathrm{Hom}(N, P))$, and the flatness criteria of 2.19. For the Tor exercises (24–26) I assume the standard facts: Tor_n is symmetric, is computed from a free resolution of either variable, and has long exact sequences in each variable.

This chapter covers the three in-text exercises (2.2, 2.15, 2.20) and the 28 end-of-chapter exercises. Exercises 14 and 23 as printed only introduce definitions (direct limits, tensor product of a family of algebras) and contain nothing to prove; I note this and verify the small points the text leaves to the reader.

In-text exercises

Exercise 2.2

> **Problem.** Let M, N, P be submodules of an A-module. Show that (i) $\mathrm{Ann}(M+N) = \mathrm{Ann}(M) \cap \mathrm{Ann}(N)$; (ii) $(N : P) = \mathrm{Ann}\big((N+P)/N\big)$.

Solution. (i) If $a(M+N) = 0$ then $aM = 0$ and $aN = 0$ because $M, N \subseteq M+N$. Conversely if $aM = aN = 0$ then $a(m+n) = am + an = 0$.

(ii) $a \in (N : P) \iff aP \subseteq N$. On the other hand $a \in \mathrm{Ann}((N+P)/N) \iff ax \in N$ for all $x \in N+P$. Writing $x = n + p$ we have $ax = an + ap$ with $an \in N$ automatically, so this is equivalent to $ap \in N$ for all $p \in P$. □

Exercise 2.15

> **Problem.** Let A, B be rings, M an A-module, P a B-module and N an (A, B)-bimodule (an A-module and a B-module with $a(xb) = (ax)b$). Then $M \otimes_A N$ is naturally a B-module, $N \otimes_B P$ an A-module, and $(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$.

Solution. *Module structures.* By compatibility, multiplication $y \mapsto by$ on N is A-linear, so $1 \otimes (b\cdot) : M \otimes_A N \to M \otimes_A N$ is well defined; this gives $b(x \otimes y) = x \otimes by$, and the module

axioms hold on generators $x \otimes y$ and hence everywhere. Likewise $y \mapsto ay$ is B-linear, so $N \otimes_B P$ is an A-module via $a(y \otimes z) = ay \otimes z$.

Isomorphism (same method as 2.14). Fix $z \in P$. The map $(x, y) \mapsto x \otimes (y \otimes z)$ of $M \times N$ into $M \otimes_A (N \otimes_B P)$ is A-bilinear, hence gives $f_z : M \otimes_A N \to M \otimes_A (N \otimes_B P)$ with $f_z(x \otimes y) = x \otimes (y \otimes z)$. Make $M \otimes_A (N \otimes_B P)$ a B-module via $b(x \otimes w) = x \otimes bw$ (multiplication by b on $N \otimes_B P$, $y \otimes z \mapsto by \otimes z$, is A-linear by compatibility, as in the first paragraph). The map $(t, z) \mapsto f_z(t)$ is B-bilinear: it is additive in each variable, and $f_{bz}(x \otimes y) = x \otimes (y \otimes bz) = x \otimes (by \otimes z) = f_z(x \otimes by) = f_z(b(x \otimes y))$, so $f_{bz}(t) = f_z(bt)$ for all t (both sides are additive in t); also $f_z(bt) = bf_z(t)$ on generators, so f_z is B-linear and $f_{bz} = bf_z$. It induces $f : (M \otimes_A N) \otimes_B P \to M \otimes_A (N \otimes_B P)$, $(x \otimes y) \otimes z \mapsto x \otimes (y \otimes z)$.

Conversely, fix $x \in M$; the map $(y, z) \mapsto (x \otimes y) \otimes z$ of $N \times P$ into $(M \otimes_A N) \otimes_B P$ is B-bilinear and B-balanced since $(x \otimes by) \otimes z = b(x \otimes y) \otimes z = (x \otimes y) \otimes bz$; it gives $\gamma_x : N \otimes_B P \to (M \otimes_A N) \otimes_B P$. Then $(x, t) \mapsto \gamma_x(t)$ is A-bilinear, giving $g : M \otimes_A (N \otimes_B P) \to (M \otimes_A N) \otimes_B P$ with $g(x \otimes (y \otimes z)) = (x \otimes y) \otimes z$.

$f \circ g$ and $g \circ f$ are the identity on the generators $x \otimes (y \otimes z)$, resp. $(x \otimes y) \otimes z$, so f, g are inverse isomorphisms. □

Exercise 2.20

> **Problem.** If $f : A \to B$ is a ring homomorphism and M is a flat A-module, then $M_B = B \otimes_A M$ is a flat B-module. (Use the canonical isomorphisms 2.14, 2.15.)

Solution. Let N be a B-module. Using 2.15 (with B as a (B, A)-bimodule) and 2.14(iv),

$$N \otimes_B M_B = N \otimes_B (B \otimes_A M) \cong (N \otimes_B B) \otimes_A M \cong N \otimes_A M,$$

where in the last term N is regarded as an A-module by restriction of scalars. These isomorphisms are natural in N. Let $u : N' \to N$ be an injective homomorphism of B-modules. It is also an injective A-linear map, so $u \otimes 1 : N' \otimes_A M \to N \otimes_A M$ is injective because M is A-flat. By naturality, this map corresponds to $u \otimes 1 : N' \otimes_B M_B \to N \otimes_B M_B$, which is therefore injective. By 2.19(iii), M_B is a flat B-module. □

End-of-chapter exercises

Exercise 1

> **Problem.** Show that $(\mathbf{Z}/m\mathbf{Z}) \otimes_{\mathbf{Z}} (\mathbf{Z}/n\mathbf{Z}) = 0$ if m, n are coprime.

Solution. Write $1 = am + bn$ with $a, b \in \mathbf{Z}$. For $x \in \mathbf{Z}/m\mathbf{Z}$, $y \in \mathbf{Z}/n\mathbf{Z}$ we have $mx = 0$ and $ny = 0$, so

$$x \otimes y = (am + bn)(x \otimes y) = a\,(mx) \otimes y + b\,x \otimes (ny) = 0.$$

These elements generate the tensor product, so it is 0. □

Exercise 2

Problem. Let A be a ring, $\mathfrak{a}$ an ideal, M an A-module. Show that $(A/\mathfrak{a}) \otimes_A M \cong M/\mathfrak{a}M$. [Tensor the exact sequence $0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0$ with M.]

Solution. By right exactness (2.18), $\mathfrak{a} \otimes M \to A \otimes M \to (A/\mathfrak{a}) \otimes M \to 0$ is exact. Under $A \otimes M \cong M$ $(a \otimes x \mapsto ax)$, the image of $\mathfrak{a} \otimes M$ is the set of sums $\sum a_i x_i$ with $a_i \in \mathfrak{a}$, i.e. $\mathfrak{a}M$. Hence $(A/\mathfrak{a}) \otimes M \cong M/\mathfrak{a}M$ (via $\bar{a} \otimes x \mapsto \overline{ax}$). □

Exercise 3

Problem. Let A be a local ring, M and N finitely generated A-modules. Prove that if $M \otimes N = 0$ then $M = 0$ or $N = 0$. [Let $\mathfrak{m}$ be the maximal ideal, $k = A/\mathfrak{m}$, $M_k = k \otimes_A M \cong M/\mathfrak{m}M$ by Exercise 2. By Nakayama's lemma, $M_k = 0 \Rightarrow M = 0$. But $M \otimes_A N = 0 \Rightarrow (M \otimes_A N)_k = 0 \Rightarrow M_k \otimes_k N_k = 0 \Rightarrow M_k = 0$ or $N_k = 0$, since M_k, N_k are vector spaces over a field.]

Solution. Let $\mathfrak{m}$ be the maximal ideal and $k = A/\mathfrak{m}$. By Exercise 2, $M_k = k \otimes_A M \cong M/\mathfrak{m}M$.

Nakayama: M is finitely generated, so $M_k = 0$ means $\mathfrak{m}M = M$ and hence $M = 0$ by 2.6 (as $\mathfrak{m} = \mathfrak{R}$ for a local ring). Likewise for N.

Base change of the tensor product: $M_k \otimes_k N_k \cong (M \otimes_A N)_k$. Indeed, using 2.15 (with k as a (k, A)-bimodule) and 2.14,

$$M_k \otimes_k N_k = M_k \otimes_k (k \otimes_A N) \cong (M_k \otimes_k k) \otimes_A N \cong M_k \otimes_A N = (k \otimes_A M) \otimes_A N \cong k \otimes_A (M \otimes_A N).$$

Now suppose $M \otimes_A N = 0$. Then $M_k \otimes_k N_k = 0$. For vector spaces over a field, $\dim(V \otimes W) = \dim V \cdot \dim W$, so $M_k = 0$ or $N_k = 0$, hence $M = 0$ or $N = 0$. □

Exercise 4

Problem. Let M_i $(i \in I)$ be any family of A-modules, and let M be their direct sum. Prove that M is flat $\iff$ each M_i is flat.

Solution. Tensor product commutes with arbitrary direct sums: for every module P, $\mathrm{Hom}\big((\bigoplus M_i) \otimes N, P\big) \cong \mathrm{Hom}\big(\bigoplus M_i, \mathrm{Hom}(N, P)\big) \cong \prod_i \mathrm{Hom}(M_i, \mathrm{Hom}(N, P)) \cong \prod_i \mathrm{Hom}(M_i \otimes N, P) \cong \mathrm{Hom}\big(\bigoplus (M_i \otimes N), P\big)$, so $M \otimes N \cong \bigoplus_i (M_i \otimes N)$, naturally in N.

Let $u : N' \to N$ be injective. Under these isomorphisms $1_M \otimes u$ corresponds to $\bigoplus_i (1_{M_i} \otimes u)$. A direct sum of homomorphisms is injective iff each summand is. Hence $1_M \otimes u$ is injective for all injective u iff each $1_{M_i} \otimes u$ is. By 2.19(iii), M is flat iff every M_i is flat. □

Exercise 5

Problem. Let $A[x]$ be the ring of polynomials in one indeterminate over a ring A. Prove that $A[x]$ is a flat A-algebra. [Use Exercise 4.]

Solution. As an A-module, $A[x] = \bigoplus_{n\geq 0} Ax^n$ is a direct sum of copies of A. A is flat since $A \otimes N \cong N$ naturally, so tensoring with A preserves injections. By Exercise 4, $A[x]$ is flat. $\square$

Exercise 6

> **Problem.** For any A-module M, let $M[x]$ denote the set of all polynomials $m_0 + m_1 x + \cdots + m_r x^r$ ($m_i \in M$). Defining the product of an element of $A[x]$ and an element of $M[x]$ in the obvious way, show that $M[x]$ is an $A[x]$-module. Show that $M[x] \cong A[x] \otimes_A M$.

Solution. *Module structure.* Put $\big(\sum a_i x^i\big)\big(\sum m_j x^j\big) = \sum_k \big(\sum_{i+j=k} a_i m_j\big) x^k$. The axioms are the same computations as for polynomial multiplication, using the A-module axioms of M.

Isomorphism. The map $A[x] \times M \to M[x]$, $\big(\sum a_i x^i, m\big) \mapsto \sum (a_i m) x^i$ is A-bilinear, so it induces an A-linear $\varphi : A[x] \otimes_A M \to M[x]$ with $\varphi(f \otimes m) = fm$. It is $A[x]$-linear: $\varphi(g(f \otimes m)) = gfm = g\varphi(f \otimes m)$. Define $\psi : M[x] \to A[x] \otimes_A M$ by $\psi\big(\sum m_i x^i\big) = \sum x^i \otimes m_i$; it is additive. Then $\varphi\psi\big(\sum m_i x^i\big) = \sum m_i x^i$, and on generators $ax^i \otimes m$ of the tensor product, $\psi\varphi(ax^i \otimes m) = \psi((am)x^i) = x^i \otimes am = ax^i \otimes m$. So φ and ψ are inverse isomorphisms. $\square$

Exercise 7

> **Problem.** Let $\mathfrak{p}$ be a prime ideal in A. Show that $\mathfrak{p}[x]$ is a prime ideal in $A[x]$. If $\mathfrak{m}$ is a maximal ideal in A, is $\mathfrak{m}[x]$ a maximal ideal in $A[x]$?

Solution. $\mathfrak{p}[x]$ (polynomials with all coefficients in $\mathfrak{p}$) is the kernel of the surjective homomorphism $A[x] \to (A/\mathfrak{p})[x]$ (reduce coefficients). Since $A/\mathfrak{p}$ is a domain, so is $(A/\mathfrak{p})[x]$; thus $\mathfrak{p}[x]$ is prime.

For $\mathfrak{m}$ maximal, $A[x]/\mathfrak{m}[x] \cong (A/\mathfrak{m})[x]$ is a polynomial ring over a field, which is not a field (x is not a unit). So $\mathfrak{m}[x]$ is prime but *not* maximal: $\mathfrak{m}[x] \subsetneq \mathfrak{m}[x] + (x) \subsetneq A[x]$. $\square$

Exercise 8

> **Problem.** (i) If M and N are flat A-modules, then so is $M \otimes_A N$. (ii) If B is a flat A-algebra and N is a flat B-module, then N is flat as an A-module.

Solution. (i) Let $u : P' \to P$ be injective. Then $1_N \otimes u : N \otimes P' \to N \otimes P$ is injective (N flat), and then $1_M \otimes (1_N \otimes u)$ is injective (M flat). By the natural associativity isomorphism $(M \otimes N) \otimes P \cong M \otimes (N \otimes P)$ (2.14(ii)), this is $1_{M\otimes N} \otimes u$. So $M \otimes N$ is flat.

(ii) For an A-module P, $N \otimes_A P \cong N \otimes_B (B \otimes_A P)$, naturally in P (2.15 and 2.14(iv): $N \otimes_B (B \otimes_A P) \cong (N \otimes_B B) \otimes_A P \cong N \otimes_A P$). Let $u : P' \to P$ be an injective A-linear map. Since B is A-flat, $1_B \otimes u : B \otimes_A P' \to B \otimes_A P$ is injective, and it is B-linear. Since N is B-flat, $1_N \otimes (1_B \otimes u)$ is injective. By the natural isomorphism this map is $1_N \otimes_A u$. Hence N is A-flat. $\square$

Exercise 9

Problem. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules. If M' and M'' are finitely generated, then so is M.

Solution. Let $0 \to M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$ be exact. Let $x'_1, \dots, x'_r$ generate M' and $y''_1, \dots, y''_s$ generate M''; choose $y_j \in M$ with $v(y_j) = y''_j$. We claim the $u(x'_i)$ and y_j generate M. Let $z \in M$ and write $v(z) = \sum a_j y''_j = v(\sum a_j y_j)$. Then $z - \sum a_j y_j \in \operatorname{Ker} v = \operatorname{Im} u$, so $z - \sum a_j y_j = u(\sum b_i x'_i) = \sum b_i u(x'_i)$. □

Exercise 10

Problem. Let A be a ring, $\mathfrak{a}$ an ideal contained in the Jacobson radical of A; let M be an A-module and N a finitely generated A-module, and let $u : M \to N$ be a homomorphism. If the induced homomorphism $M/\mathfrak{a}M \to N/\mathfrak{a}N$ is surjective, then u is surjective.

Solution. Surjectivity of $M/\mathfrak{a}M \to N/\mathfrak{a}N$ means $N = u(M) + \mathfrak{a}N$. Put $Q = N/u(M)$, a finitely generated module. Then $\mathfrak{a}Q = (\mathfrak{a}N + u(M))/u(M) = N/u(M) = Q$. Since $\mathfrak{a} \subseteq \mathfrak{R}$, Nakayama's lemma (2.6) gives $Q = 0$, i.e. u is surjective. □

Exercise 11

Problem. Let A be a ring $\neq 0$. Show that $A^m \cong A^n \Rightarrow m = n$. [Let $\mathfrak{m}$ be a maximal ideal of A and let $\phi : A^m \to A^n$ be an isomorphism. Then $1 \otimes \phi : (A/\mathfrak{m}) \otimes A^m \to (A/\mathfrak{m}) \otimes A^n$ is an isomorphism between vector spaces of dimensions m and n over $k = A/\mathfrak{m}$. Hence $m = n$.] (Cf. Chapter 3, Exercise 15.) If $\phi : A^m \to A^n$ is surjective, then $m \geq n$. If $\phi : A^m \to A^n$ is injective, is it always the case that $m \leq n$?

Solution. Let $\mathfrak{m}$ be a maximal ideal and $k = A/\mathfrak{m}$. Tensor product commutes with finite direct sums and $k \otimes_A A \cong k$, so $k \otimes_A A^m \cong k^m$. If $\phi : A^m \to A^n$ is an isomorphism, $1 \otimes \phi : k^m \to k^n$ is an isomorphism of k-vector spaces, so $m = n$.

If ϕ is surjective, $1 \otimes \phi : k^m \to k^n$ is surjective (right exactness), so $m \geq n$.

If $\phi : A^m \to A^n$ is injective, then yes, $m \leq n$ always. Suppose $m > n$ and let $\iota : A^n \to A^m$ be the inclusion onto the first n coordinates, $W = \iota(A^n)$. Then $\psi = \iota \circ \phi : A^m \to A^m$ is injective with $\operatorname{Im} \psi \subseteq W$. By 2.4 (Cayley–Hamilton, with $\mathfrak{a} = (1)$) ψ satisfies a monic equation $r(\psi) = 0$ of degree $m \geq 1$. Claim: if a monic r of degree $k \geq 1$ satisfies $r(\psi) = 0$, then $r = ts$ with s monic of degree $k - 1$ and $s(\psi) = 0$. Indeed write $r(t) = t\, s(t) + d$ with $d \in A$ and s monic. Apply $r(\psi) = 0$ to the basis vector e_m:

$$\psi\big(s(\psi)e_m\big) + d\, e_m = 0.$$

The first term lies in $\operatorname{Im} \psi \subseteq W$, so $d\, e_m \in W$; as $m > n$, the m-th coordinate of elements of W is 0, so $d = 0$. Then $\psi \circ s(\psi) = 0$ and, ψ being injective, $s(\psi) = 0$. Repeating m times we reach a monic polynomial of degree 0, i.e. $1 = 0$ as an endomorphism of A^m. Thus $A^m = 0$, so $A = 0$, contradiction. Hence $m \leq n$. □

Exercise 12

Problem. Let M be a finitely generated A-module and $\phi : M \to A^n$ a surjective homomorphism. Show that $\mathrm{Ker}(\phi)$ is finitely generated. [Let $e_1, \ldots, e_n$ be a basis of A^n and choose $u_i \in M$ such that $\phi(u_i) = e_i$. Show that M is the direct sum of $\mathrm{Ker}(\phi)$ and the submodule generated by $u_1, \ldots, u_n$.]

Solution. Let U be the submodule generated by $u_1, \ldots, u_n$ and $K = \operatorname{Ker}\phi$. For $x = \sum a_i u_i \in U$ we have $\phi(x) = (a_1, \ldots, a_n)$; so $\phi|_U : U \to A^n$ is an isomorphism (surjective, and injective since $\phi(x) = 0$ forces all $a_i = 0$). Consequently $K \cap U = 0$. For $z \in M$ write $\phi(z) = \sum a_i e_i = \phi(\sum a_i u_i)$; then $z - \sum a_i u_i \in K$, so $M = K + U$. Hence $M = K \oplus U$ and $K \cong M/U$ is a quotient of the finitely generated module M, so K is finitely generated. □

Exercise 13

Problem. Let $f : A \to B$ be a ring homomorphism and let N be a B-module. Regarding N as an A-module by restriction of scalars, form the B-module $N_B = B \otimes_A N$. Show that the homomorphism $g : N \to N_B$ which maps y to $1 \otimes y$ is injective and that $g(N)$ is a direct summand of N_B. [Define $p : N_B \to N$ by $p(b \otimes y) = by$, and show that $N_B = \mathrm{Im}(g) \oplus \mathrm{Ker}(p)$.]

Solution. The map $B \times N \to N$, $(b, y) \mapsto by$ is A-bilinear (since $(f(a)b)y = b(f(a)y)$), so $p : N_B \to N$, $p(b \otimes y) = by$ is a well-defined B-linear map. Also g is A-linear ($g(ay) = 1 \otimes f(a)y = f(a) \otimes y = a(1 \otimes y)$), and $p \circ g(y) = p(1 \otimes y) = y$. Hence g is injective.

Moreover $N_B = \operatorname{Im} g \oplus \operatorname{Ker} p$: any $z \in N_B$ is $z = gp(z) + \big(z - gp(z)\big)$ with $p(z - gp(z)) = p(z) - p(z) = 0$, and if $g(y) \in \operatorname{Ker} p$ then $y = pg(y) = 0$. So $g(N)$ is a direct summand of N_B (as A-modules; $\operatorname{Ker} p$ is even a B-submodule). □

Exercise 14

Problem. (Direct limits.) A partially ordered set I is *directed* if for each pair i, j in I there exists $k \in I$ such that $i \leq k$ and $j \leq k$. Let A be a ring, I a directed set and $(M_i)_{i \in I}$ a family of A-modules indexed by I. For each pair $i \leq j$ let $\mu_{ij} : M_i \to M_j$ be an A-homomorphism, and suppose that

(1) μ_{ii} is the identity mapping of M_i, for all $i \in I$;

(2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ whenever $i \leq j \leq k$.

Then the modules M_i and homomorphisms μ_{ij} form a *direct system* $\mathbf{M} = (M_i, \mu_{ij})$ over I. The *direct limit* $M = \varinjlim M_i$ is constructed as follows: let C be the direct sum of the M_i, identify each M_i with its canonical image in C, let D be the submodule of C generated by all elements $x_i - \mu_{ij}(x_i)$ with $i \leq j$ and $x_i \in M_i$, let $M = C/D$, let $\mu : C \to M$ be the projection and μ_i its restriction to M_i. From the construction it is clear that $\mu_i = \mu_j \circ \mu_{ij}$ whenever $i \leq j$.

Solution. As printed, this exercise only sets up the definition of the direct limit; there is nothing to prove. For completeness, the closing assertion: for $x_i \in M_i$ and $i \leq j$, $x_i - \mu_{ij}(x_i) \in D$, so $\mu_i(x_i) = \mu(x_i) = \mu(\mu_{ij}(x_i)) = \mu_j \mu_{ij}(x_i)$. □

Exercise 15

Problem. In the situation of Exercise 14, show that every element of M can be written in the form $\mu_i(x_i)$ for some $i \in I$ and some $x_i \in M_i$. Show also that if $\mu_i(x_i) = 0$ then there exists $j \geq i$ such that $\mu_{ij}(x_i) = 0$ in M_j.

Solution. *First claim.* An element of $M = C/D$ is the class of a finite sum $x_{i_1} + \cdots + x_{i_r}$ with $x_{i_s} \in M_{i_s}$. Since I is directed there is $j \geq i_1, \ldots, i_r$. Modulo D, $x_{i_s} \equiv \mu_{i_s j}(x_{i_s})$, so the class equals $\mu_j\big(\sum_s \mu_{i_s j}(x_{i_s})\big)$.

Second claim. Suppose $\mu_i(x_i) = 0$, i.e. $x_i \in D$. Then $x_i = \sum_{s=1}^{r}\big(y_s - \mu_{k_s l_s}(y_s)\big)$ with $y_s \in M_{k_s}$, $k_s \leq l_s$. Choose $j \geq i$ and $j \geq k_s, l_s$ for all s. Let $C_j = \bigoplus_{k \leq j} M_k \subseteq C$ and define the homomorphism $\theta : C_j \to M_j$ by $\theta|_{M_k} = \mu_{kj}$. This relation takes place in C_j, and $\theta\big(y_s - \mu_{k_s l_s}(y_s)\big) = \mu_{k_s j}(y_s) - \mu_{l_s j}\mu_{k_s l_s}(y_s) = 0$ by axiom (2). Therefore $\mu_{ij}(x_i) = \theta(x_i) = 0$. □

Exercise 16

Problem. Show that the direct limit is characterized (up to isomorphism) by the following property. Let N be an A-module and for each $i \in I$ let $\alpha_i : M_i \to N$ be an A-module homomorphism such that $\alpha_i = \alpha_j \circ \mu_{ij}$ whenever $i \leq j$. Then there exists a unique homomorphism $\alpha : M \to N$ such that $\alpha_i = \alpha \circ \mu_i$ for all $i \in I$.

Solution. *Existence.* Define $C \to N$ by $(x_i) \mapsto \sum \alpha_i(x_i)$. It kills the generators $x_i - \mu_{ij}(x_i)$ of D, since $\alpha_i(x_i) - \alpha_j\mu_{ij}(x_i) = 0$. So it factors through $M = C/D$, giving α with $\alpha\mu_i = \alpha_i$.

Uniqueness. By Exercise 15 every element of M is some $\mu_i(x_i)$, and $\alpha(\mu_i(x_i)) = \alpha_i(x_i)$ is forced.

Characterization. Let (M', μ_i') be another module with homomorphisms $\mu_i' : M_i \to M'$ satisfying $\mu_i' = \mu_j'\mu_{ij}$ and the same universal property. Applying the property of M to (M', μ_i') gives $\alpha : M \to M'$ with $\alpha\mu_i = \mu_i'$; applying the property of M' to (M, μ_i) gives $\beta : M' \to M$ with $\beta\mu_i' = \mu_i$. Then $\beta\alpha\mu_i = \mu_i = 1_M\mu_i$ for all i, so $\beta\alpha = 1_M$ by uniqueness; similarly $\alpha\beta = 1_{M'}$. Hence $M \cong M'$ by a unique isomorphism compatible with the maps. □

Exercise 17

Problem. Let $(M_i)_{i \in I}$ be a family of submodules of an A-module, such that for each pair of indices i, j in I there exists $k \in I$ such that $M_i + M_j \subseteq M_k$. Define $i \leq j$ to mean $M_i \subseteq M_j$ and let $\mu_{ij} : M_i \to M_j$ be the embedding of M_i in M_j. Show that $\varinjlim M_i = \sum M_i = \bigcup M_i$. In particular, any A-module is the direct limit of its finitely generated submodules.

Solution. Let M be the ambient module. (If $M_i = M_j$ for some $i \neq j$, then $\leq$ is only a preorder; the construction of Exercise 14 and Exercises 15, 16 go through verbatim for directed preorders.) I is directed: given i, j take k with $M_i + M_j \subseteq M_k$, so $i, j \leq k$. The set $\sum M_i$ equals $\bigcup M_i$: a finite sum $x_{i_1} + \cdots + x_{i_r}$ lies in a single M_k with $k \geq i_1, \ldots, i_r$ (induction using directedness).

The inclusions $\iota_i : M_i \to \sum M_i$ satisfy $\iota_i = \iota_j\mu_{ij}$, so by Exercise 16 there is $\alpha : \varinjlim M_i \to \sum M_i$ with $\alpha\mu_i = \iota_i$. It is surjective since its image contains every M_i. It is injective: an element of the limit is $\mu_i(x_i)$ (Exercise 15), and $\alpha(\mu_i(x_i)) = x_i$; if this is 0 then $x_i = 0$ in M_i, so $\mu_i(x_i) = 0$. Hence $\varinjlim M_i \cong \sum M_i = \bigcup M_i$.

For the last statement take I to be the set of finitely generated submodules of a module M, ordered by inclusion: the sum of two finitely generated submodules is finitely generated, and every $x \in M$ lies in Ax. So $M = \bigcup M_i = \varinjlim M_i$. □

Exercise 18

> **Problem.** Let $\mathbf{M} = (M_i, \mu_{ij})$, $\mathbf{N} = (N_i, \nu_{ij})$ be direct systems of A-modules over the same directed set. Let M, N be the direct limits and $\mu_i : M_i \to M$, $\nu_i : N_i \to N$ the associated homomorphisms. A *homomorphism* $\Phi : \mathbf{M} \to \mathbf{N}$ is by definition a family of A-module homomorphisms $\phi_i : M_i \to N_i$ such that $\phi_j \circ \mu_{ij} = \nu_{ij} \circ \phi_i$ whenever $i \leq j$. Show that Φ defines a unique homomorphism $\phi = \varinjlim \phi_i : M \to N$ such that $\phi \circ \mu_i = \nu_i \circ \phi_i$ for all $i \in I$.

Solution. Put $\alpha_i = \nu_i \circ \phi_i : M_i \to N$. For $i \leq j$: $\alpha_j \mu_{ij} = \nu_j \phi_j \mu_{ij} = \nu_j \nu_{ij} \phi_i = \nu_i \phi_i = \alpha_i$. By Exercise 16 there is a unique $\phi : M \to N$ with $\phi \mu_i = \alpha_i = \nu_i \phi_i$. □

Exercise 19

> **Problem.** A sequence of direct systems and homomorphisms $\mathbf{M} \to \mathbf{N} \to \mathbf{P}$ is *exact* if the corresponding sequence of modules and module homomorphisms is exact for each $i \in I$. Show that the sequence $M \to N \to P$ of direct limits is then exact. [Use Exercise 15.]

Solution. Let $\Phi = (\phi_i) : \mathbf{M} \to \mathbf{N}$, $\Psi = (\psi_i) : \mathbf{N} \to \mathbf{P}$, with limits ϕ, ψ, and write μ_i, ν_i, ρ_i for the maps into M, N, P and ν_{ij}, π_{ij} for the transition maps of $\mathbf{N}, \mathbf{P}$.

$\psi\phi = 0$: $\psi\phi\mu_i = \psi\nu_i\phi_i = \rho_i\psi_i\phi_i = 0$ for all i, and M is generated by the images of the μ_i (Exercise 15), so $\psi\phi = 0$.

$\operatorname{Ker}\psi \subseteq \operatorname{Im}\phi$: let $y \in N$ with $\psi(y) = 0$. By Exercise 15, $y = \nu_i(y_i)$ for some i and $y_i \in N_i$. Then $0 = \psi(y) = \rho_i\psi_i(y_i)$, so by Exercise 15 there is $j \geq i$ with $\pi_{ij}\psi_i(y_i) = 0$, i.e. $\psi_j(\nu_{ij}(y_i)) = 0$. By exactness at N_j, $\nu_{ij}(y_i) = \phi_j(x_j)$ for some $x_j \in M_j$. Then $y = \nu_j\nu_{ij}(y_i) = \nu_j\phi_j(x_j) = \phi\mu_j(x_j) \in \operatorname{Im}\phi$. □

Exercise 20

> **Problem.** (Tensor products commute with direct limits.) Keeping the same notation as in Exercise 14, let N be any A-module. Then $(M_i \otimes N, \mu_{ij} \otimes 1)$ is a direct system; let $P = \varinjlim(M_i \otimes N)$ be its direct limit. For each $i \in I$ we have a homomorphism $\mu_i \otimes 1 : M_i \otimes N \to M \otimes N$, hence by Exercise 16 a homomorphism $\psi : P \to M \otimes N$. Show that ψ is an isomorphism, so that $\varinjlim(M_i \otimes N) \cong (\varinjlim M_i) \otimes N$. [For each $i \in I$, let $g_i : M_i \times N \to M_i \otimes N$ be the canonical bilinear mapping. Passing to the limit we obtain a mapping $g : M \times N \to P$. Show that g is A-bilinear and hence defines a homomorphism $\phi : M \otimes N \to P$. Verify that $\phi \circ \psi$ and $\psi \circ \phi$ are identity mappings.]

Solution. Let $\lambda_i : M_i \otimes N \to P$ be the limit maps. The maps $\mu_i \otimes 1$ satisfy $(\mu_j \otimes 1)(\mu_{ij} \otimes 1) = \mu_i \otimes 1$, so ψ exists with $\psi\lambda_i = \mu_i \otimes 1$.

Construction of g. Two preliminary facts. (a) Any two (or finitely many) elements of M can be written $\mu_k(x), \mu_k(x')$ with the same index k (Exercise 15 and directedness: move both

up to a common k using $\mu_i = \mu_k\mu_{ik}$). (b) If $\mu_i(x_i) = \mu_j(x_j)$, then there is $l \geq i, j$ with $\mu_{il}(x_i) = \mu_{jl}(x_j)$. Indeed take $k \geq i, j$; then $\mu_k(\mu_{ik}x_i - \mu_{jk}x_j) = 0$, so by Exercise 15 there is $l \geq k$ with $\mu_{kl}(\mu_{ik}x_i - \mu_{jk}x_j) = 0$.

Define $g(\mu_i(x), y) = \lambda_i(x \otimes y)$. It is well defined: if $\mu_i(x_i) = \mu_j(x_j)$, take l as in (b); then $\lambda_i(x_i \otimes y) = \lambda_l(\mu_{il}x_i \otimes y) = \lambda_l(\mu_{jl}x_j \otimes y) = \lambda_j(x_j \otimes y)$. By (a) and the bilinearity of $\otimes$ in M_k, g is additive in each variable and A-linear: $g(\mu_k(x) + \mu_k(x'), y) = \lambda_k((x + x') \otimes y) = g(\mu_k(x), y) + g(\mu_k(x'), y)$, and similarly for y and scalars. So g is A-bilinear and induces $\phi : M \otimes N \to P$ with $\phi(\mu_i(x) \otimes y) = \lambda_i(x \otimes y)$.

Inverse isomorphisms. $\phi\psi\lambda_i(x \otimes y) = \phi(\mu_i(x) \otimes y) = \lambda_i(x \otimes y)$; the elements $\lambda_i(x \otimes y)$ generate P (Exercise 15), so $\phi\psi = 1_P$. And $\psi\phi(\mu_i(x) \otimes y) = \psi\lambda_i(x \otimes y) = \mu_i(x) \otimes y$; these elements generate $M \otimes N$ (Exercise 15), so $\psi\phi = 1$. □

Exercise 21

> **Problem.** Let $(A_i)_{i \in I}$ be a family of rings indexed by a directed set I, and for each pair $i \leq j$ in I let $\alpha_{ij} : A_i \to A_j$ be a ring homomorphism, satisfying conditions (1) and (2) of Exercise 14. Regarding each A_i as a $\mathbf{Z}$-module we can then form the direct limit $A = \varinjlim A_i$. Show that A inherits a ring structure from the A_i so that the mappings $A_i \to A$ are ring homomorphisms. The ring A is the *direct limit* of the system (A_i, α_{ij}). If $A = 0$ prove that $A_i = 0$ for some $i \in I$. [Remember that all rings have identity elements!]

Solution. Let $\alpha_i : A_i \to A$ be the canonical maps (additive, with $\alpha_i = \alpha_j\alpha_{ij}$).

Multiplication. Given $a, a' \in A$, write $a = \alpha_k(x)$, $a' = \alpha_k(x')$ with a common index k (Exercise 15) and define $aa' = \alpha_k(xx')$. This is well defined: if also $a = \alpha_l(y)$, $a' = \alpha_l(y')$, use Exercise 15 (as in Exercise 20(b)) to find $m \geq k, l$ with $\alpha_{km}(x) = \alpha_{lm}(y)$ and $\alpha_{km}(x') = \alpha_{lm}(y')$ (find indices for each equation and pass to a larger one; the equalities persist since the α_{jm} are compatible). Then

$$\alpha_k(xx') = \alpha_m(\alpha_{km}(x)\alpha_{km}(x')) = \alpha_m(\alpha_{lm}(y)\alpha_{lm}(y')) = \alpha_l(yy').$$

Ring axioms. Any finite number of elements of A are of the form $\alpha_k(\cdot)$ for one k, and α_k is additive and multiplicative by definition; so associativity, commutativity and distributivity hold in A because they hold in A_k. The element $1 = \alpha_i(1_{A_i})$ is independent of i (as $\alpha_{ij}(1) = 1$, $\alpha_i = \alpha_j\alpha_{ij}$, and any two indices have a common upper bound), and is an identity. By construction each α_i is a ring homomorphism.

If $A = 0$: then $\alpha_i(1_{A_i}) = 1_A = 0$ for some (any) i, so by Exercise 15 there is $j \geq i$ with $\alpha_{ij}(1) = 0$. But $\alpha_{ij}(1) = 1_{A_j}$, so $1 = 0$ in A_j, i.e. $A_j = 0$. □

Exercise 22

> **Problem.** Let (A_i, α_{ij}) be a direct system of rings and let $\mathfrak{N}_i$ be the nilradical of A_i. Show that $\varinjlim \mathfrak{N}_i$ is the nilradical of $\varinjlim A_i$. If each A_i is an integral domain, then $\varinjlim A_i$ is an integral domain.

Solution. Let $A = \varinjlim A_i$, with $\alpha_i : A_i \to A$. A ring homomorphism maps nilpotents to nilpotents, so $\alpha_{ij}(\mathfrak{N}_i) \subseteq \mathfrak{N}_j$ and $(\mathfrak{N}_i)$ is a direct system of modules; the inclusions $\mathfrak{N}_i \to A_i$ induce $\varinjlim \mathfrak{N}_i \to A$.

Injective. An element of $\varinjlim \mathfrak{N}_i$ is $\lambda_i(x)$ with $x \in \mathfrak{N}_i$ (Exercise 15). If it maps to $\alpha_i(x) = 0$, then $\alpha_{ij}(x) = 0$ for some j (Exercise 15), so $\lambda_i(x) = \lambda_j(\alpha_{ij}x) = 0$. (This is also Exercise 19 applied to $0 \to \mathfrak{N} \to A$.)

Image is $\mathfrak{N}(A)$. If $x \in \mathfrak{N}_i$, $x^n = 0$, then $\alpha_i(x)^n = 0$, so $\alpha_i(x) \in \mathfrak{N}(A)$. Conversely if $a = \alpha_i(x) \in \mathfrak{N}(A)$, $a^n = 0$, then $\alpha_i(x^n) = 0$, so by Exercise 15 $\alpha_{ij}(x)^n = \alpha_{ij}(x^n) = 0$ for some $j \geq i$; thus $\alpha_{ij}(x) \in \mathfrak{N}_j$ and $a = \alpha_j(\alpha_{ij}x)$ is in the image of $\mathfrak{N}_j$. Hence $\varinjlim \mathfrak{N}_i \cong \mathfrak{N}(A)$.

Domains. Let each A_i be a domain (so $A_i \neq 0$). By Exercise 21, $A \neq 0$. If $ab = 0$ in A, write $a = \alpha_k(x)$, $b = \alpha_k(y)$; then $\alpha_k(xy) = 0$, so $\alpha_{kj}(x)\alpha_{kj}(y) = \alpha_{kj}(xy) = 0$ for some $j \geq k$. As A_j is a domain, $\alpha_{kj}(x) = 0$ or $\alpha_{kj}(y) = 0$, hence $a = 0$ or $b = 0$. □

Exercise 23

> **Problem.** Let $(B_\lambda)_{\lambda\in\Lambda}$ be a family of A-algebras. For each finite subset J of Λ let B_J denote the tensor product (over A) of the B_λ for $\lambda \in J$. If J' is another finite subset of Λ and $J \subseteq J'$, there is a canonical A-algebra homomorphism $B_J \to B_{J'}$. Let B denote the direct limit of the rings B_J as J runs through all finite subsets of Λ. The ring B has a natural A-algebra structure for which the homomorphisms $B_J \to B$ are A-algebra homomorphisms. The A-algebra B is the *tensor product* of the family $(B_\lambda)_{\lambda\in\Lambda}$.

Solution. As printed, this exercise only defines the tensor product of an arbitrary family of algebras; there is nothing to prove. The statements the text leaves to the reader are routine to check: the homomorphism $B_J \to B_{J'}$ $(J \subseteq J')$ is $\bigotimes_{\lambda\in J} x_\lambda \mapsto \bigotimes_{\lambda\in J'} y_\lambda$ with $y_\lambda = x_\lambda$ for $\lambda \in J$ and $y_\lambda = 1$ otherwise (well defined by the universal property of the multi-tensor product 2.12*, and an A-algebra homomorphism because it is multiplicative and sends 1 to 1); the finite subsets of Λ form a directed set under inclusion $(J, J' \leq J \cup J')$, and the transition maps satisfy conditions (1), (2) of Exercise 14. So B is a ring by Exercise 21; the structure maps $A \to B_J$ are compatible with the transition maps, hence define a ring homomorphism $A \to B$ for which each $B_J \to B$ is an A-algebra homomorphism. □

Exercise 24

> **Problem.** If M is an A-module, the following are equivalent: (i) M is flat; (ii) $\operatorname{Tor}_n^A(M, N) = 0$ for all $n > 0$ and all A-modules N; (iii) $\operatorname{Tor}_1^A(M, N) = 0$ for all A-modules N.

Solution. (i)⇒(ii). Take a free resolution $\cdots \to F_2 \to F_1 \to F_0 \to N \to 0$ of N (exact). Since M is flat, $\cdots \to M \otimes F_2 \to M \otimes F_1 \to M \otimes F_0 \to M \otimes N \to 0$ is exact. Its homology at $M \otimes F_n$ $(n > 0)$ is $\operatorname{Tor}_n(M, N) = 0$.

(ii)⇒(iii). Trivial.

(iii)⇒(i). Let $0 \to N' \to N \to N'' \to 0$ be exact. The Tor long exact sequence contains

$$\operatorname{Tor}_1(M, N'') \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0.$$

Since $\operatorname{Tor}_1(M, N'') = 0$, the map $M \otimes N' \to M \otimes N$ is injective. So $M \otimes -$ preserves injections, i.e. M is flat (2.19). □

Exercise 25

Problem. Let $0 \to N' \to N \to N'' \to 0$ be an exact sequence, with N'' flat. Then N' is flat $\iff$ N is flat. [Use Exercise 24 and the Tor exact sequence.]

Solution. For any module L the Tor long exact sequence (in the second variable) gives

$$\mathrm{Tor}_2(L, N'') \to \mathrm{Tor}_1(L, N') \to \mathrm{Tor}_1(L, N) \to \mathrm{Tor}_1(L, N'').$$

As N'' is flat, $\mathrm{Tor}_n(L, N'') \cong \mathrm{Tor}_n(N'', L) = 0$ for $n \geq 1$ (Exercise 24 and symmetry of Tor). Hence $\mathrm{Tor}_1(L, N') \cong \mathrm{Tor}_1(L, N)$ for every L. By Exercise 24 and symmetry, N' is flat $\iff$ $\mathrm{Tor}_1(N', L) = 0$ for all L $\iff$ $\mathrm{Tor}_1(N, L) = 0$ for all L $\iff$ N is flat. $\square$

Exercise 26

Problem. Let N be an A-module. Then N is flat $\iff$ $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated ideals $\mathfrak{a}$ in A. [Show first that N is flat if $\mathrm{Tor}_1(M, N) = 0$ for all finitely generated A-modules M, by using (2.19). If M is finitely generated, let $x_1, \ldots, x_n$ be a set of generators of M, and let M_i be the submodule generated by $x_1, \ldots, x_i$. By considering the successive quotients M_i/M_{i-1} and using Exercise 25, deduce that N is flat if $\mathrm{Tor}_1(M, N) = 0$ for all *cyclic* A-modules M, i.e., all M generated by a single element, and therefore of the form $A/\mathfrak{a}$ for some ideal $\mathfrak{a}$. Finally use (2.19) again to reduce to the case where $\mathfrak{a}$ is a finitely generated ideal.]

Solution. ($\Rightarrow$) is Exercise 24 (with symmetry of Tor), since N is flat.

($\Leftarrow$) Assume $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated ideals $\mathfrak{a}$.

Step 1: $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ *for every ideal* $\mathfrak{a}$. The sequence $0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0$ gives the exact sequence $0 = \mathrm{Tor}_1(A, N) \to \mathrm{Tor}_1(A/\mathfrak{a}, N) \to \mathfrak{a} \otimes N \to A \otimes N = N$, so $\mathrm{Tor}_1(A/\mathfrak{a}, N) = \mathrm{Ker}(\mathfrak{a} \otimes N \to N)$. Let $z = \sum_{i=1}^{r} x_i \otimes y_i \in \mathfrak{a} \otimes N$ map to $\sum x_i y_i = 0$. Let $\mathfrak{a}_0 = (x_1, \ldots, x_r) \subseteq \mathfrak{a}$ (finitely generated) and $z_0 = \sum x_i \otimes y_i \in \mathfrak{a}_0 \otimes N$. Then z_0 maps to 0 in N, i.e. $z_0 \in \mathrm{Ker}(\mathfrak{a}_0 \otimes N \to N) = \mathrm{Tor}_1(A/\mathfrak{a}_0, N) = 0$. So $z_0 = 0$ and hence its image z in $\mathfrak{a} \otimes N$ is 0.

Step 2: $\mathrm{Tor}_1(M, N) = 0$ *for every finitely generated* M. Let M be generated by $x_1, \ldots, x_n$, $M_i = \sum_{j \leq i} Ax_j$, $M_0 = 0$. Each M_i/M_{i-1} is cyclic, $\cong A/\mathfrak{a}$, so $\mathrm{Tor}_1(M_i/M_{i-1}, N) = 0$ by Step 1. From the exact sequence $\mathrm{Tor}_1(M_{i-1}, N) \to \mathrm{Tor}_1(M_i, N) \to \mathrm{Tor}_1(M_i/M_{i-1}, N)$ and induction on i we get $\mathrm{Tor}_1(M_i, N) = 0$; for $i = n$ this is $\mathrm{Tor}_1(M, N) = 0$.

Step 3: N *is flat.* Let $u : M' \to M$ be injective with M, M' finitely generated. From $0 \to M' \to M \to M/M' \to 0$ we get $\mathrm{Tor}_1(M/M', N) \to M' \otimes N \to M \otimes N$ exact, and $\mathrm{Tor}_1(M/M', N) = 0$ by Step 2 (M/M' is finitely generated). So $u \otimes 1$ is injective, and N is flat by 2.19(iv). $\square$

Exercise 27

Problem. A ring A is *absolutely flat* if every A-module is flat. Prove that the following are equivalent:

(i) A is absolutely flat.

(ii) Every principal ideal is idempotent.

(iii) Every finitely generated ideal is a direct summand of A.

[(i)⇒(ii). Let $x \in A$. Then $A/(x)$ is a flat A-module, hence in the commutative square with top row $\beta : (x) \otimes A \to (x) \otimes A/(x)$, vertical maps $(x) \otimes A \to A$ and $\alpha : (x) \otimes A/(x) \to A/(x)$, and bottom row $A \to A/(x)$, the mapping α is injective. Hence $\mathrm{Im}(\beta) = 0$, hence $(x) = (x^2)$. (ii)⇒(iii). Let $x \in A$. Then $x = ax^2$ for some $a \in A$, hence $e = ax$ is idempotent and we have $(e) = (x)$. Now if e, f are idempotents, then $(e, f) = (e + f - ef)$. Hence every finitely generated ideal is principal, and generated by an idempotent e, hence is a direct summand because $A = (e) \oplus (1 - e)$. (iii)⇒(i). Use the criterion of Exercise 26.]

Solution. (i)⇒(ii). Let $x \in A$. $A/(x)$ is flat, so tensoring the inclusion $(x) \hookrightarrow A$ with $A/(x)$ gives an injective map $\alpha : (x) \otimes A/(x) \to A \otimes A/(x) = A/(x)$, $y \otimes \bar{a} \mapsto \overline{ya}$. But $y \in (x)$, so $\overline{ya} = 0$: $\alpha = 0$. Hence $(x) \otimes A/(x) = 0$. By Exercise 2, $(x) \otimes A/(x) \cong (x)/(x)(x) = (x)/(x^2)$. So $(x) = (x^2)$.

(ii)⇒(iii). Let $x \in A$. Then $x \in (x^2)$, say $x = ax^2$. Put $e = ax$. Then $e^2 = a^2x^2 = a(ax^2) = ax = e$, so e is idempotent; $e \in (x)$ and $x = ax^2 = xe \in (e)$, so $(e) = (x)$. For idempotents e, f put $g = e + f - ef$. Then $ge = e + ef - ef = e$ and $gf = f$, hence $e, f \in (g) \subseteq (e, f)$, i.e. $(e, f) = (g)$; moreover $g^2 = g(e + f - ef) = e + f - ef = g$, so g is idempotent. By induction, every finitely generated ideal $(x_1, \ldots, x_n) = (e_1, \ldots, e_n)$ is generated by one idempotent e. Finally $A = (e) \oplus (1 - e)$: $x = xe + x(1 - e)$, and if $y \in (e) \cap (1 - e)$ then $y = ye = 0$ (from $y = y_1(1 - e)$ we get $ye = 0$). So (e) is a direct summand.

(iii)⇒(i). Let N be any A-module and $\mathfrak{a}$ a finitely generated ideal, $A = \mathfrak{a} \oplus \mathfrak{b}$. Then $A/\mathfrak{a} \cong \mathfrak{b}$ is a direct summand of A, hence flat (Exercise 4; A is flat). By Exercise 24, $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$. By Exercise 26, N is flat. So A is absolutely flat. □

Exercise 28

Problem. A Boolean ring is absolutely flat. The ring of Chapter 1, Exercise 7 is absolutely flat. Every homomorphic image of an absolutely flat ring is absolutely flat. If a local ring is absolutely flat, then it is a field. If A is absolutely flat, every non-unit in A is a zero-divisor.

Solution. We use Exercise 27: A is absolutely flat iff $(x) = (x^2)$ for all $x \in A$, i.e. iff for each x there is a with $x = ax^2$.

Boolean rings. $x = x^2 = 1 \cdot x^2$.

The ring of Chapter 1, Exercise 7 (every x satisfies $x^n = x$ for some $n > 1$): $x = x^n = x^{n-2}x^2$ (for $n = 2$ take $a = 1$). So it is absolutely flat.

Homomorphic images. Let $A \to B = A/\mathfrak{a}$ be surjective with A absolutely flat. Each $\bar{x} \in B$ lifts to $x \in A$ with $x = ax^2$, so $\bar{x} = \bar{a}\bar{x}^2$, hence B is absolutely flat.

Local rings. Let A be local with maximal ideal $\mathfrak{m}$ and absolutely flat. If $x \in \mathfrak{m}$, $x = ax^2$, so $x(1 - ax) = 0$. Since $ax \in \mathfrak{m}$, $1 - ax \notin \mathfrak{m}$ is a unit, so $x = 0$. Thus $\mathfrak{m} = 0$ and A is a field.

Non-units are zero-divisors. Let x be a non-unit of an absolutely flat ring $A \neq 0$, $x = ax^2$, so $x(1 - ax) = 0$. If $1 - ax = 0$ then $ax = 1$ and x would be a unit; so $1 - ax \neq 0$ and x is a zero-divisor. □